

GOLMUD, China

WMO: 528180

Lat: 36.4207N

Lon: 94.9083E

Elev: 9216

StdP: 10.42

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.5	6.2	-21.0	2.4	18.1	-18.4	2.8	19.1	14.1	24.0	12.3	23.1	2.9	270	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	83.2	54.2	80.2	52.6	77.4	51.4	57.1	76.9	55.1	73.6	53.5	72.3	4.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.0	75.8	65.1	47.7	69.3	63.8	45.3	63.4	62.9	29.0	77.0	27.4	73.4	26.2	72.7	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	12.8	11.2	-1.8	89.4	2.2	2.4	-3.4	91.1	-4.7	92.5	-5.9	93.9	-7.5	95.6		
(5)			-3.8	60.0	2.0	2.4	-5.2	61.7	-6.4	63.2	-7.6	64.5	-9.0	66.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	19.6	27.1	36.4	46.9	55.2	62.1	66.5	65.3	56.9	44.3	31.2	21.1		
	DBStd	17.29	5.47	6.38	6.49	6.63	5.36	4.70	4.92	4.51	4.73	5.53	5.91	4.92		
	HDD50	3813	943	640	425	134	17	1	0	0	4	189	563	897		
	HDD65	7666	1408	1060	888	543	305	107	41	51	247	641	1013	1362		
	CDD50	1796	0	0	2	41	178	364	512	475	210	14	0	0		
	CDD65	172	0	0	0	0	1	20	87	61	3	0	0	0		
	CDH74	1509	0	0	0	9	32	191	743	493	40	1	0	0		
	CDH80	280	0	0	0	2	1	17	167	91	2	0	0	0		
(14) Wind	WSAvg	4.9	4.0	4.6	5.3	5.9	6.1	5.8	5.5	5.1	4.6	4.2	3.9	3.8		
(15) Precipitation	PrecAvg	1.6	0.0	0.0	0.0	0.0	0.2	0.3	0.4	0.3	0.2	0.0	0.0	0.0		
	PrecMax	4.0	0.1	0.1	0.3	0.5	0.9	1.1	1.7	1.3	0.6	0.6	0.2	0.2		
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.7	0.0	0.0	0.1	0.1	0.2	0.2	0.4	0.3	0.2	0.1	0.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	53.0	64.8	74.1	77.9	82.4	88.3	86.4	78.6	69.1	54.3	42.3		
		MCWB	27.4	32.8	38.9	44.0	47.2	52.3	57.3	55.6	52.2	43.3	34.4	28.2		
	2%	DB	37.4	47.7	57.4	68.5	73.7	78.7	84.3	82.4	73.9	63.4	50.2	37.5		
		MCWB	24.5	30.2	35.3	41.9	45.3	50.7	55.1	54.0	49.2	40.4	32.2	25.1		
	5%	DB	33.8	43.4	53.3	64.4	70.5	75.9	81.3	79.2	70.6	59.6	46.5	34.1		
		MCWB	22.5	27.9	33.4	39.9	44.1	49.3	53.4	52.6	47.5	38.8	30.4	23.1		
	10%	DB	30.3	39.4	49.6	60.5	67.6	73.4	78.2	76.3	67.7	56.2	43.2	31.1		
		MCWB	20.5	25.9	31.5	38.1	42.8	48.2	52.4	51.3	45.8	37.2	28.7	21.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	33.2	39.4	45.0	48.6	56.0	60.7	59.5	56.1	44.8	34.8	28.5		
		MCDB	41.5	52.0	63.5	69.8	70.8	74.1	83.3	80.9	71.8	63.6	53.4	42.3		
	2%	WB	24.8	30.4	35.7	42.6	46.9	53.1	57.6	56.9	52.1	41.8	32.3	25.2		
		MCDB	36.7	46.4	55.6	66.6	68.5	71.6	78.6	75.5	68.0	60.0	49.1	36.7		
	5%	WB	22.8	28.3	33.7	40.5	45.4	51.4	55.5	54.9	49.7	39.7	30.6	23.3		
		MCDB	33.4	42.9	52.6	62.8	66.6	69.9	75.2	72.7	66.6	57.5	45.9	33.7		
	10%	WB	20.7	26.1	31.8	38.7	43.9	50.0	54.0	53.1	47.6	37.9	28.8	21.5		
		MCDB	30.0	39.1	49.0	59.2	64.6	68.7	73.5	71.4	64.1	55.0	42.7	30.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.9	22.3	22.8	22.5	20.9	19.1	19.9	20.0	20.8	22.5	22.1	20.6		
		MCDBR	25.0	27.1	28.7	28.0	25.4	23.4	25.1	24.9	25.0	26.8	25.4	23.6		
		MCWBR	16.3	15.9	15.2	13.2	11.4	9.9	10.0	10.5	10.9	12.6	14.1	15.2		
	5% WB	MCDBR	24.0	25.9	27.2	26.3	22.5	20.1	21.8	20.8	21.1	24.7	24.5	22.7		
		MCWBR	15.9	15.3	14.4	12.5	10.1	8.3	8.7	9.0	9.5	11.8	13.7	14.8		
(40) Clear-Sky Solar Irradiance	taub		0.229	0.271	0.336	0.395	0.408	0.406	0.397	0.384	0.350	0.296	0.248	0.226		
	taud		2.464	2.333	2.091	1.928	1.971	2.070	2.173	2.198	2.263	2.345	2.438	2.480		
	Ebn at Noon		315	309	293	280	278	278	279	281	286	295	304	310		
	Edh at Noon		26	34	47	59	58	52	47	45	39	33	26	23		
(44) All-Sky Solar Radiation	RadAvg		1030	1312	1758	2072	2235	2160	2167	2029	1744	1498	1150	961		
	RadStd		49	56	48	78	86	144	127	103	103	50	43	30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	N/A	N/A	N/A	-203	-274	+97	N/A
(47) Regional (1 neighbor)		+0.86	N/A	N/A	N/A	N/A	N/A	-229	-302	+97	N/A

Nomenclature: See separate page